Reproducibility Scorecard

A novel ensemble learning approach to extract urban impervious surface based on machine learning algorithms using SAR and optical data

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Checklist Item	What to Look For	Y	P	N
Seeds reported, many runs?	Mean ± SD, not one number			X
Splits described?	Sizes, method, grouped or time-based if needed	X		
Leakage checked?	Preprocessing inside folds; no future features			X
Tuning separated?	Validation set or nested CV		X	
Statistical test named?	Appropriate, paired when comparing models			X
Effect size and CI?	Not just a p-value			X
External validation?	New site, new period or new population	X		
Compute documented?	Hardware, runtime, versions		X	
Code and data available?	A stranger could re-run it	X		

Detailed Item Descriptions

1. **Seeds reported, many runs? (N)**

   The paper reports accuracy and performance metrics as single point values in its tables (e.g., 92% overall accuracy for UISEM). It does not document the use of fixed random seeds to ensure exact reproducibility of stochastic machine learning algorithms, nor does it present aggregated results in a mean ± standard deviation (± SD) format across multiple independent runs.

2. **Splits described? (Y)**

The methodology clearly outlines data partitioning: an independent unseen test set of 20% was held out and excluded from training and validation. The remaining 80% was evaluated using a 5-fold cross-validation scheme, explicitly detailing both sample sizes and splitting methods.

3. **Leakage checked? (N)**

   A reference dataset of over 350,000 points was collected globally using random selection prior to cross-validation. For spatial and remote sensing data, purely random pixel sampling frequently leads to spatial data leakage due to spatial autocorrelation among adjacent pixels—an issue that is neither mentioned nor mitigated using spatial block partitioning.

4. **Tuning separated? (P)**

   The authors evaluated different feature subsets (6, 12, 18, 27, 35, and 47 features) using Forward Stepwise Selection and a correlation matrix to arrive at the optimal 27-feature set. However, a formal nested cross-validation or isolated validation set for hyperparameter tuning of individual base classifiers (e.g., XGBoost, Random Forest) is not explicitly detailed.

5. **Statistical test named? (N)**

   Performance comparisons between the proposed UISEM model, ablation baseline variants (Models 1–6), and global benchmarks (ESA, ESRI, Dynamic World) rely strictly on direct numerical metric differences. No formal statistical significance tests (e.g., McNemar's test or Wilcoxon signed-rank test) were performed to confirm that accuracy improvements were statistically significant.

6. **Effect size and CI? (N)**

   Following from the absence of formal hypothesis testing across model evaluations, neither effect sizes nor confidence intervals (CIs) are reported for the classification performance metrics.

7. **External validation? (Y)**

   The study demonstrates strong spatial external validation by testing and validating the model across **32 global cities** spanning diverse climatic zones (tropical, dry, desert, temperate, boreal, and polar). This confirms the model's generalization capability across unseen geographical environments.

8. **Compute documented? (P)**

   The paper explicitly names the cloud computing platforms used (Google Earth Engine, Google Colab, and Jupyter Notebooks). However, it omits details regarding

the exact hardware specs (CPU/GPU allocation, RAM), runtime duration for training on $350{,}000+$ samples, and specific software library versions.

9. **Code and data available? (Y)**

   The authors state that all Python scripts, Google Earth Engine (GEE) code, and generated datasets are publicly accessible on GitHub at [https://github.com/mnasarahmad/UISEM]